A Project Report on

"Customer Segmentation Using Purchase Behaviour and K-Means Clustering"

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Master of Technology

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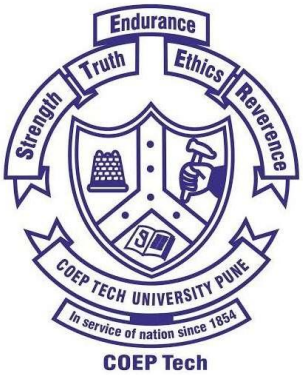
Data Science

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1. Abstract

This project segments **1,283,346 customers** of a retail business into behaviourally distinct groups using K-Means clustering. Customer-level features were engineered from transaction records (sales and returns), giving 27 columns per customer. Derived measures were validated against their formulas, redundant and heavily skewed features were identified, and the retained features were transformed with log, Yeo-Johnson and standard scaling before clustering.

Four candidate feature sets and cluster counts from K = 2 to 12 were compared using inertia, silhouette score, Davies–Bouldin index, cluster-size balance and the adjusted Rand index (ARI) across random seeds. The highest silhouette scores occurred at K = 2, which separates only active from inactive customers. K = 4 on a lean five-feature set (recency, sale frequency, monthly spending, average discount rate and average unit price) was the most reproducible choice, with a mean ARI of 0.904 across five independent training samples. The final model produced four segments: **Loyal High-Value** (456,423 customers, 35.6%), **Inactive Moderate-Spending** (359,975, 28.0%), **Discount-Driven Regulars** (206,779, 16.1%) and **Lapsed Low-Spending** (260,169, 20.3%). A full-data K-Means run reproduced the saved solution with ARI = 0.994.

The silhouette score of about 0.24 shows that the segments overlap and have soft boundaries, and the segment names are business interpretations rather than labels learned by the algorithm. The trained pipeline, scaling parameters and segment profiles were saved so that new customers can be scored consistently.

Keywords: customer segmentation, RFM, K-Means, MiniBatchKMeans, silhouette score, adjusted Rand index, retail analytics.

2. Introduction

2.1 Background

Retail businesses accumulate large volumes of transaction records, yet marketing, retention and pricing decisions are often applied to all customers alike. Customer segmentation groups customers with similar purchasing behaviour so that each group can be treated differently. Classic RFM analysis describes a customer by recency, frequency and monetary value. This project extends that idea with spending level over time, discount usage, product price level and return behaviour, and uses an unsupervised algorithm instead of manually chosen thresholds.

2.2 Problem Statement

To convert raw transaction records into customer-level behavioural features, select a defensible set of features and number of clusters for a base of more than 1.2 million customers, group the customers using K-Means, and interpret the resulting segments in business terms.

2.3 Objectives

- Prepare transaction data and build customer-level behavioural features covering recency, frequency, monetary value, discounts, product mix and returns.
- Inspect and validate the features by checking derived formulas and identifying redundancy, skewness and zero-inflation.
- Transform and scale the features so that they are suitable for distance-based clustering.

- Compare candidate feature sets and cluster counts using internal metrics and stability across random seeds.
- Fit the final K-Means model and profile each segment using interpretable median values.
- Visualise the segments and save the trained pipeline so that new customers can be scored.

2.4 Scope

The project covers a single snapshot of customer behaviour. Clustering is unsupervised, so there is no ground-truth label against which accuracy can be measured; the quality of the solution is judged with internal metrics and reproducibility. The segments describe behaviour and do not establish causes, and the segment names are interpretations added by the analyst.

3. Dataset Description

The customer-level table was built from transaction data containing sale and return invoices and stored as a parquet file. It has **1,283,346 rows (one per customer) and 27 columns**. Monetary values are in USD. The 21 numeric measures used as clustering candidates contained no missing and no infinite values, and no constant or duplicated columns.

3.1 Feature Dictionary

Feature	Definition / Formula	Unit / Range
Customer ID	Unique identifier for each customer	ID
Last_Purchase_Date	MAX(Date) across all transactions (sale + return)	Date
first_Purchase_Date	MIN(Date) across all transactions	Date
Recency	analysis_date − MAX(Date) for that customer	Days
Frequency	Count of distinct Invoice IDs (sale + return)	Count of orders
Sale_Frequency	Count of distinct Invoice IDs where Transaction Type = Sale	Count
Unique_Products	Count of distinct Product IDs purchased or returned	Count
Net_Monetary	SUM(Line Total) over sale and return rows; returns are negative	USD
Gross_Monetary	SUM(Line Total) over Sale rows only	USD
Customer_Lifetime	MAX(Date) − MIN(Date)	Days
Average_Purchase_Interval	Customer_Lifetime ÷ (Frequency − 1); 0 if Frequency = 1	Days
Total_Quantity	SUM(Quantity) over Sale rows	Units
Average_Unit_Price	MEAN(Unit Price) over Sale rows	USD
average_discount_rate	MEAN(Discount) over Sale rows	Ratio, 0–1
products_with_base_price	SUM(Quantity) where Discount = 0 and type = Sale	Units
products_with_discount	SUM(Quantity) where Discount > 0 and type = Sale	Units
Return_Count	Count of distinct Invoice IDs where type = Return	Count
Returned_Quantity	SUM(Quantity) over Return rows	Units
Return_Value	ABS(SUM(Line Total)) over Return rows	USD

Feature	Definition / Formula	Unit / Range
Favorite_Shopping_Day_in_week	Mode of weekday name over Sale rows	Text
Favorite_Shopping_month_in_year	Mode of month name over Sale rows	Text
Monthly_Spending_Mean	Sale Line Total summed per month, then averaged across months	USD
Monthly_Spending_Std	Standard deviation of the monthly totals; 0 if one active month	USD
Spending_Trend	Slope of a linear fit (np.polyfit, degree 1) through monthly totals; 0 if one active month	USD per month
Is_One_Time_Customer	1 if Frequency = 1, else 0	Binary
average_order_value	Gross_Monetary ÷ Sale_Frequency	USD
Return_Rate	Return_Count ÷ Sale_Frequency	Ratio

3.2 Summary Statistics

Table below summarises the main numeric features. Most monetary and activity measures are right-skewed, with means above medians, and the typical customer is a modest spender who purchases a few times.

Feature	Mean	Std. dev.	Min	Median	Max
Recency (days)	218.75	194.29	1	141	808
Sale_Frequency	3.38	2.63	1	3	35
Unique_Products	4.73	4.14	1	3	59
Net_Monetary (USD)	220.95	213.37	0	154.22	3,054.11
Customer_Lifetime (days)	313.12	258.43	0	323	807
Average_Unit_Price (USD)	51.00	17.72	3.18	49.20	161.49
average_discount_rate	0.126	0.143	0	0.088	0.60
Monthly_Spending_Mean (USD)	74.19	50.95	1.78	62.10	1,040.90
Return_Rate	0.047	0.147	0	0	1.00
Spending_Trend (USD/month)	−0.53	54.36	−992.18	0	1,051.75

4. Methodology

4.1 Workflow

The project followed this pipeline: loading the customer feature table → inspection → formula validation → redundancy, variance and distribution analysis → feature shortlisting → transformation and scaling → comparison of feature sets and cluster counts → stability testing → final K-Means fit → segment profiling and naming → visual validation → saving the model artefacts.

4.2 Tools and Technologies

Python 3, pandas and NumPy (data handling), Matplotlib, Seaborn and Plotly (visualisation), scikit-learn (preprocessing, K-Means, MiniBatchKMeans, PCA and metrics), joblib and pickle (model persistence), Parquet (data storage) and Google Colab (development).

4.3 Data Validation

Every derived feature was recomputed from its definition and compared with the stored value. All checks matched on 100% of the rows tested, which confirms that the feature table is internally consistent.

Feature	**Expected relationship**	**Rows checked**	**Match**
average_order_value	Gross_Monetary ÷ Sale_Frequency	1,283,346	100%
Average_Purchase_Interval	Customer_Lifetime ÷ (Frequency − 1)	963,482	100%
Return_Rate	Return_Count ÷ Sale_Frequency	1,283,346	100%
Net_Monetary	Gross_Monetary − Return_Value	1,283,346	100% (max error 4.5×10^{-13})
Total_Quantity	products_with_base_price + products_with_discount	1,283,346	100% (max error 0)

4.4 Redundancy and Distribution Analysis

Identifiers, date columns, the two categorical favourite-shopping columns and the binary Is_One_Time_Customer flag were excluded from the numeric candidate matrix, leaving 21 candidates. Spearman and Pearson correlation matrices were computed, and 22 feature pairs with an absolute Spearman correlation of at least 0.85 were flagged as redundant. The strongest were Returned_Quantity and Return_Value (0.998), Return_Count and Returned_Quantity (0.997), Frequency and Sale_Frequency (0.988), Unique_Products and Total_Quantity (0.980), and Net_Monetary and Gross_Monetary (0.979).

Variance, dominant-value share, zero share, skewness and IQR-based outlier counts were also examined. These checks were used as screening flags and not as automatic deletion rules. The four return features are zero for 85.7% of customers, so they carry little information for separating most customers.

4.5 Feature Selection and Preprocessing

Twelve features were shortlisted from the 21 candidates, taking one or two representatives from each behavioural group (recency, activity, value, product, discount, return and spending behaviour) and avoiding near-duplicates. Because the features are strongly skewed and clustering

is distance-based, each group was transformed as shown below, giving a 12-column scaled matrix with no NaN or infinite values.

Group	Features	Transformation
Skewed positive features (9)	Recency, Sale_Frequency, Customer_Lifetime, Net_Monetary, Unique_Products, Average_Unit_Price, Monthly_Spending_Mean, Monthly_Spending_Std, products_with_discount	log(1 + x), then StandardScaler
Trend (1)	Spending_Trend (positive and negative values)	Yeo–Johnson power transform, then StandardScaler
Rates (2)	average_discount_rate, Return_Rate	StandardScaler

4.6 Models and Evaluation

For experiments, MiniBatchKMeans was trained on random samples of 30,000 to 100,000 customers and evaluated on samples of 3,000 to 6,000 customers, with random seeds fixed for reproducibility. Models were compared using **inertia (distortion)**, the **silhouette score** (higher is better), the **Davies–Bouldin index** (lower is better), the **smallest and largest cluster share**, and the **adjusted Rand index (ARI)** between solutions from different seeds or samples, which measures how reproducible the segments are. The final model is a standard K-Means (n_init = 10, random_state = 42) fitted on all customers.

5. Exploratory Data Analysis (Outputs)

5.1 Distributions

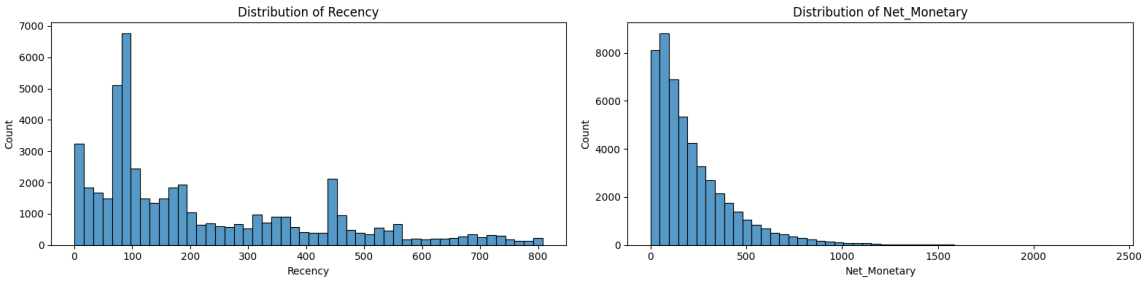


Figure 1: Distributions of Recency and Net_Monetary (50,000-customer sample)

Recency ranges from 1 to 808 days with a median of 141 and a mean of 219. The histogram is uneven, with noticeable peaks near 100 days and again near 450 days, rather than a smooth decay. Net_Monetary is strongly right-skewed (skewness 2.07): the median customer has net spend of about $154, while a small tail of customers spends above $1,000, up to $3,054. About a quarter of customers (roughly 26%, derived from the segment-level one-time shares) bought only once.

5.2 Skewness and Zero-Inflation

The features with the highest skewness are listed below. Return features are the most extreme because most customers never return anything; the spending measures are also strongly right-skewed, which is why log and Yeo–Johnson transformations were applied before clustering.

Feature	Skewness	Zero share	Median
Return_Value	5.859	85.7%	0
Returned_Quantity	5.054	85.7%	0
Return_Rate	4.346	85.7%	0
average_order_value	3.028	—	57.50
Monthly_Spending_Mean	2.720	—	62.10
Return_Count	2.717	85.7%	0
products_with_discount	2.275	—	1
Monthly_Spending_Std	2.225	—	24.95
Gross_Monetary / Net_Monetary	2.072	0% / 0.9%	162.97 / 154.22

Spending_Trend is balanced around zero: 473,013 customers have a negative trend, 350,795 have exactly zero (mostly customers active in a single month) and 459,538 have a positive trend. Net_Monetary is zero for 11,321 customers, whose purchases were fully offset by returns.

5.3 Correlation Analysis

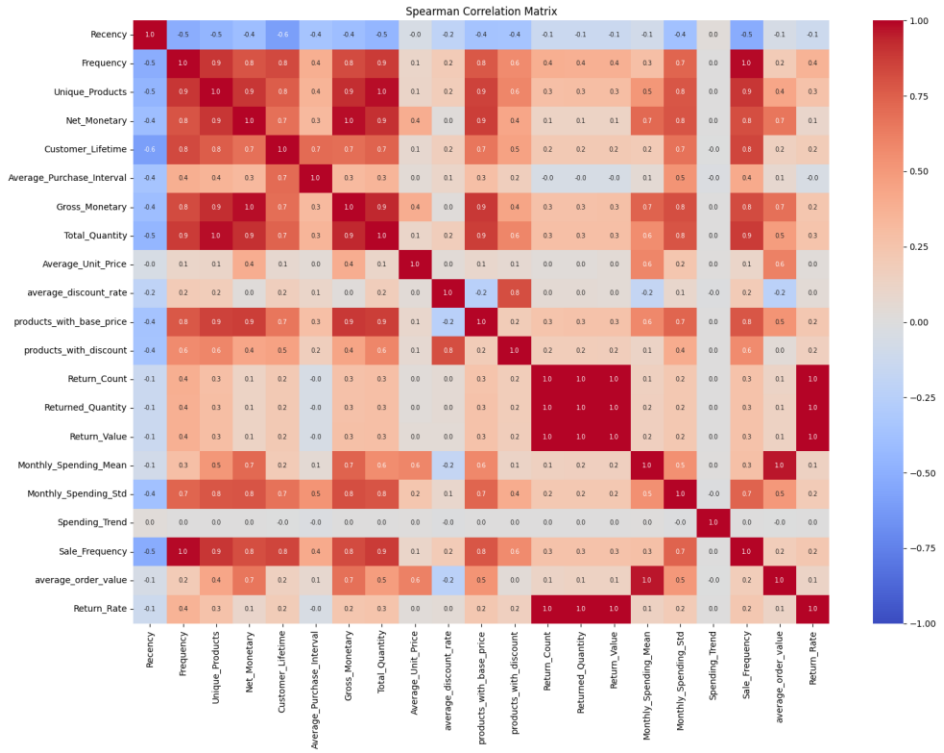


Figure 2: Spearman correlation matrix of the 21 numeric candidate features

The heatmap shows three tight blocks of related features: the return features (correlations above 0.99), the activity and value features (Frequency, Unique_Products, Total_Quantity, Net_Monetary and Gross_Monetary, with correlations of roughly 0.8 to 0.98) and the monthly spending features. Recency is negatively related to activity (about −0.5 with Frequency), meaning that customers who bought recently also tend to buy more often. Average_Unit_Price is only weakly related to the other measures (Spearman 0.10 with Frequency), so it adds information that the volume-based features do not carry.

6. Model Development and Results

6.1 Feature-Set Comparison

Four feature sets were tested. The first three were compared first using MiniBatchKMeans for K = 2 to 8 on the same 30,000-customer sample; the lean set was introduced after the stability results below.

Feature set	Features
A: Baseline 12	All 12 shortlisted scaled features
B: Core 6	Recency, Sale_Frequency, Net_Monetary, Unique_Products, average_discount_rate, Return_Rate
C: Extended 10	Core 6 plus Customer_Lifetime, Monthly_Spending_Mean, Monthly_Spending_Std, Spending_Trend
D: Lean 5 (final)	Recency, Sale_Frequency, Monthly_Spending_Mean, average_discount_rate, Average_Unit_Price

The three best silhouette results for each of the first three sets were:

Feature set	K	Silhouette	Davies–Bouldin	Smallest cluster	Largest cluster
A: Baseline 12	2	0.2794	1.4292	42.5%	57.5%
A: Baseline 12	3	0.1970	1.7169	27.4%	38.9%
A: Baseline 12	6	0.1932	1.6167	1.3%	35.4%
B: Core 6	4	0.3282	1.0360	3.6%	45.9%
B: Core 6	3	0.3111	1.2804	18.1%	44.3%
B: Core 6	2	0.3071	1.2450	45.6%	54.4%
C: Extended 10	2	0.3449	1.2255	34.3%	65.7%
C: Extended 10	5	0.2544	1.3981	7.7%	35.6%
C: Extended 10	6	0.2255	1.5434	3.2%	34.9%

The baseline 12-feature set gave the weakest separation. Core 6 with K = 4 had the best silhouette and Davies–Bouldin values of the K = 4 candidates, and Extended 10 with K = 2 scored highest overall, so these two were carried forward for a reproducibility check.

6.2 Stability across Random Seeds

Each candidate was refitted with three seeds (21, 42 and 99) and the resulting labels on the same evaluation customers were compared using ARI. Core 6 with K = 4 was unreliable: pairwise ARI ranged from 0.27 to 0.55 (mean 0.44), and the smallest cluster varied between 2.6% and 10.8% of customers. Extended 10 with K = 2 was highly stable (mean ARI 0.96) but only separates active customers from inactive ones: 839,776 customers (65%) with median recency of 93 days and four purchases against 443,570 customers (35%) with median recency of 354 days and a single purchase. Such a split is too coarse to guide different marketing actions.

Stability on the Extended 10 features was then checked for K = 2 to 5:

K	Mean silhouette	Mean ARI	Interpretation
2	0.3533	0.9625	Stable but only active vs inactive
3	0.2252	0.8022	Fairly stable
4	0.2451	0.4012	Unstable on these features
5	0.2405	0.6565	Moderately stable

Four segments are the smallest number that gives separate business actions, but on the Extended 10 features K = 4 was the least reproducible. This pointed to the feature set rather than to K as the problem, and motivated a leaner set with fewer correlated and zero-heavy inputs.

6.3 Final Feature Set and Choice of K

The lean set uses one feature from each of five behavioural dimensions: recency, activity (Sale_Frequency), spending level (Monthly_Spending_Mean), discount behaviour (average_discount_rate) and product price level (Average_Unit_Price). Customer_Lifetime and Net_Monetary were left out because they are strongly correlated with frequency and spending, and the return features were left out because 85.7% of customers have no returns; returns are still used to profile the segments. K was swept from 2 to 12 on a 6,000-customer sample.

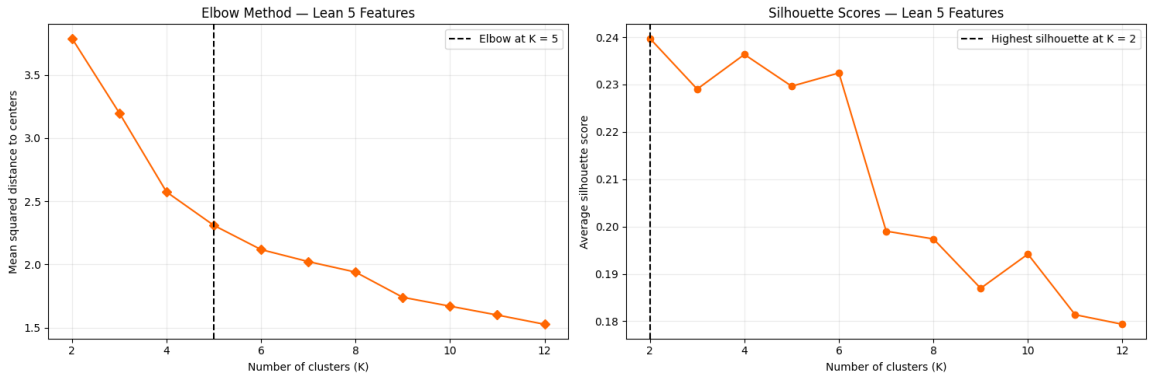


Figure 3: Elbow curve and silhouette scores for K = 2 to 12 (lean five-feature set)

K	Distortion	Silhouette
2	3.7901	0.2397
3	3.1999	0.2290
4	2.5740	0.2363
5	2.3096	0.2296
6	2.1177	0.2324
7	2.0224	0.1990
8	1.9392	0.1974
9	1.7396	0.1870
10	1.6693	0.1942
11	1.5991	0.1814
12	1.5259	0.1794

The elbow is estimated at K = 5 and the highest silhouette at K = 2, but the silhouette is almost flat from K = 2 to K = 6 (0.229 to 0.240), so it does not separate the options clearly. The decision was therefore based on reproducibility. Five models were trained on independent 100,000-customer samples (seeds 21, 42, 99, 123 and 2026) and compared on the same 5,000 evaluation customers:

K	Mean silhouette	Mean ARI	Minimum ARI
3	0.227	0.525	0.310
4	0.249	0.904	0.825
5	0.225	0.609	0.471

K = 4 is clearly the most reproducible choice (mean ARI 0.904), and it also gives four actionable groups. A second test restricted to repeat customers and replacing average unit price with Spending_Trend was less stable (mean ARI 0.43 to 0.50 for K = 3 to 5) and was not adopted.

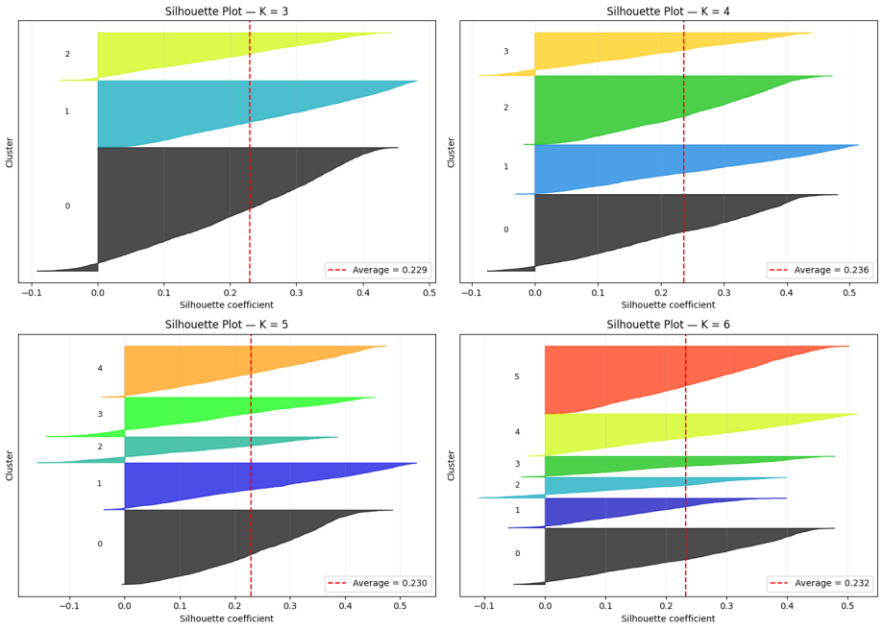


Figure 4: Silhouette plots for K = 3, 4, 5 and 6 (lean five-feature set)

6.4 Final Model and Validation

The final model is K-Means with K = 4, n_init = 10 and random_state = 42, fitted on the scaled lean features of all 1,283,346 customers. To check that the result does not depend on the sampling used during experimentation, it was compared with an independent full-data K-Means run (n_init = 5):

Check	**Result**
ARI between final solution and independent full-data K-Means	0.994
Inertia of the two solutions	3,285,206 vs 3,285,214 (gap ≈ 0.00%)
ARI between full-data K-Means runs with seeds 1, 2 and 3	0.993, 0.994 and 0.999
Silhouette score at K = 4 (6,000-customer sample)	0.236

These results show that the final clustering is reproducible. The silhouette of about 0.24 is moderate, which is typical for behavioural data where customers form a continuum rather than isolated clusters, so the segment boundaries should be read as soft.

7. Segment Profiling and Interpretation

7.1 Median Profile of Each Segment

Segments were profiled with medians of the original, untransformed features because the spending measures are strongly skewed. Cluster numbers in the figures correspond to the segments

as 0 = Loyal High-Value, 1 = Inactive Moderate-Spending, 2 = Discount-Driven Regulars and 3 = Lapsed Low-Spending.

Segment	Customers	Share	Recency (days)	Sale frequency	Monthly spend ($)	Discount rate	Unit price ($)
0: Loyal High-Value	456,423	35.6%	80	5	71.75	0.12	50.67
1: Inactive Moderate-Spending	359,975	28.0%	279	2	91.75	0.00	59.52
2: Discount-Driven Regulars	206,779	16.1%	164	2	38.96	0.37	47.85
3: Lapsed Low-Spending	260,169	20.3%	279	2	35.97	0.00	33.40

Supporting measures that were not used for clustering confirm the differences between the segments:

Segment	Net spend ($)	Unique products	Lifetime (days)	One-time buyers	Ever returned	Rising spend trend
0: Loyal High-Value	322.65	7	578	0.5%	24.0%	48.8%
1: Inactive Moderate-Spending	166.32	3	146	34.8%	10.9%	31.6%
2: Discount-Driven Regulars	66.36	2	90	41.2%	8.1%	27.2%
3: Lapsed Low-Spending	50.55	2	43	47.0%	7.0%	25.7%

7.2 Segment Descriptions

Loyal High-Value (Cluster 0). The largest segment, with the lowest median recency (80 days), the highest sale frequency (5), the highest net spend ($322.65) and the longest relationship (578 days). Almost none are one-time buyers (0.5%), and 48.8% show rising monthly spending. This segment also has the highest share of customers who have returned a product (24.0%), consistent with greater purchasing volume.

Inactive Moderate-Spending Customers (Cluster 1). Customers who have not bought for a long time (median recency 279 days) but who spent well when they were active: the highest monthly spending ($91.75) and average unit price ($59.52) of any segment, with almost no discounted purchases. Their relationship was short (median lifetime 146 days), and 34.8% bought only once.

Discount-Driven Regulars (Cluster 2). The only segment with a high discount rate (median 0.37), and 45.0% of these customers have an average discount of 0.40 or more, against at most 0.2% elsewhere. They spend little per month ($38.96) and have a short lifetime (90 days). Their favourite shopping month is strongly seasonal: 65.0% have December as their favourite month, compared with 33.3% for the Loyal segment and about 15% for the two inactive segments.

Lapsed Low-Spending Customers (Cluster 3). Customers with long inactivity (median recency 279 days), the lowest net spend ($50.55), the lowest average unit price ($33.40) and the shortest lifetime (43 days). Nearly half (47.0%) bought only once, and they rarely use discounts.

Clusters 1 and 3 have the same median recency and purchase frequency, so they are separated mainly by how much they spend and how expensive their products are. Favourite weekday is similar in all four segments (Saturday and Sunday are the most common, at roughly 21% to 26% each), so it does not distinguish them. The segment names were revised after profiling: the first labels for Clusters 1 and 3 implied that the groups were mostly one-time buyers, but only 34.8% and 47.0% are, so names describing inactivity and spending level were adopted.

7.3 Suggested Business Use

The following actions are suggestions derived from the profiles and have not been tested on campaign data.

Segment	Possible action
Loyal High-Value	Loyalty rewards, early access and retention offers; monitor the 24% who have returned products and the segment's declining-spend customers.
Inactive Moderate-Spending	Personalised win-back campaigns that feature higher-priced items, since these customers spent most per month before becoming inactive.
Discount-Driven Regulars	Time promotions to seasonal peaks such as December and keep deep discounts targeted here instead of across the whole base.
Lapsed Low-Spending	Low-cost reactivation (for example email) with limited spend per customer, since their historical value is the lowest.

8. Outputs

8.1 Segment Visualisations

The following figures were produced to check that the segments are interpretable and visually consistent with the numeric profiles.



Figure 5: Customer distribution across clusters

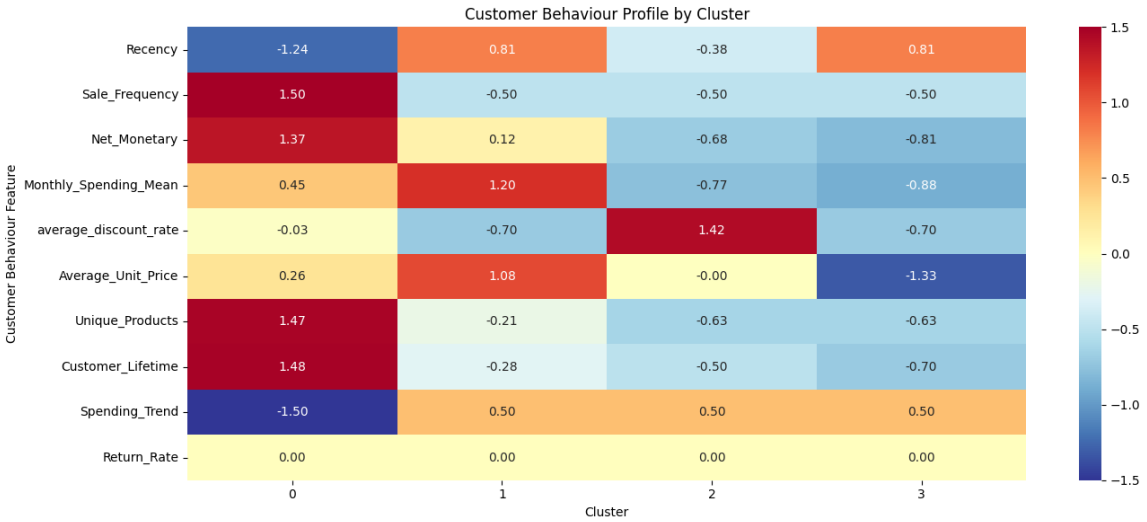


Figure 6: Standardised median behaviour profile by cluster (heatmap)

The heatmap makes the contrasts visible: Cluster 0 is high on frequency, product variety and lifetime and low on recency; Cluster 2 stands out on discount rate; Cluster 1 stands out on unit price and monthly spending; and Cluster 3 is low on almost every measure.

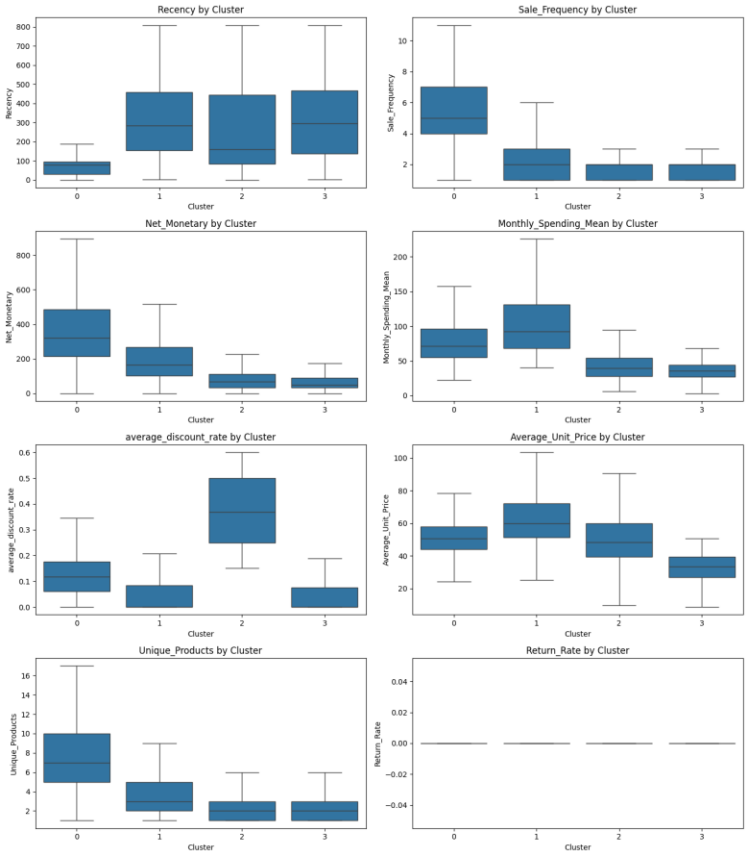


Figure 7: Box plots of key behaviour features by cluster (outliers hidden)

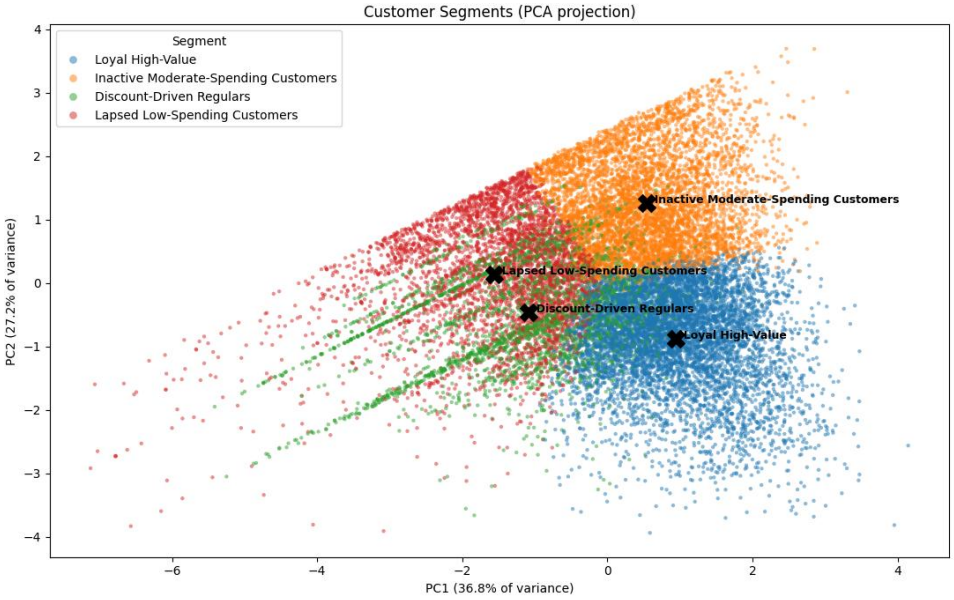


Figure 8: Customer segments in PCA space (20,000-customer sample)

The first two principal components explain 64.0% of the variance of the five clustering features (36.8% and 27.2%), and a three-component version used for an interactive 3D view explains 83.8%. The segments form a continuous cloud with visible overlap between neighbouring groups, which agrees with the moderate silhouette score.

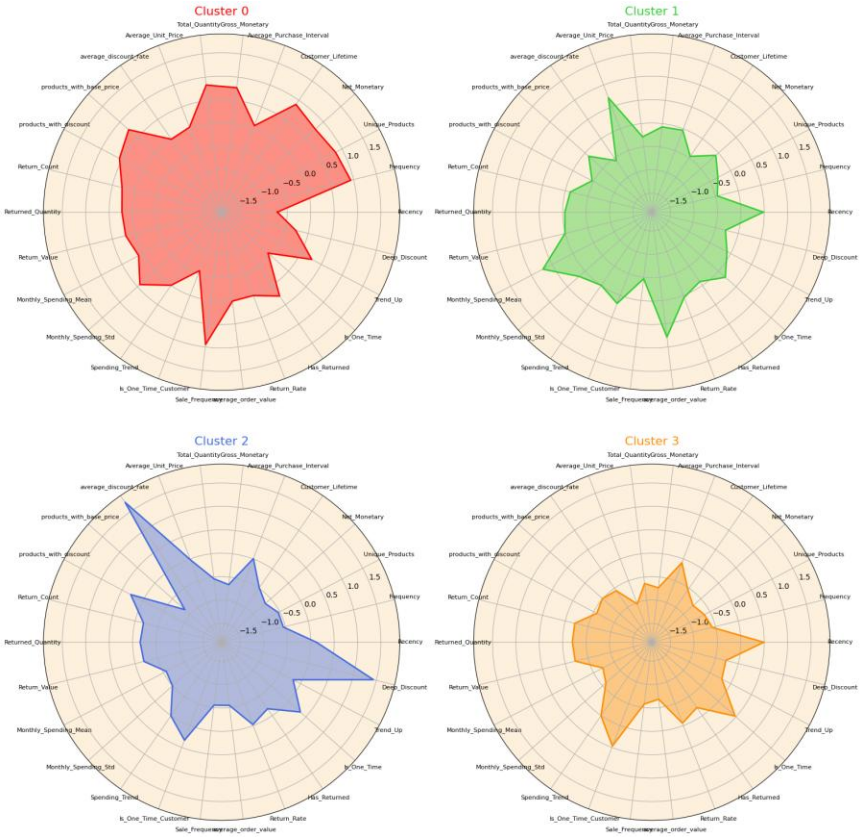


Figure 10: Radar plots of mean standardised feature values for each cluster

8.2 Saved Model Artefacts

So that new customers can be assigned to a segment with exactly the same transformations, the following objects were saved. A self-test that reloaded the artefacts and predicted segments for 5,000 randomly chosen customers reproduced the existing cluster labels with a match of 1.0 (scikit-learn 1.6.1).

Artefact	Contents
preprocessor.joblib	Fitted ColumnTransformer (log, Yeo–Johnson and standard scaling)
kmeans_k4.joblib	Final K-Means model with four clusters
model_meta.joblib	Final feature names and the names of all scaled features
artifacts.pkl	Model, five final features, scaling parameters (mean, scale, log flag), segment names, cluster medians, overall median and IQR, cluster sizes, total customers, benchmark results and analysis date

9. Limitations

Limitation	Impact
Moderate cluster separation (silhouette ≈ 0.24)	Segments overlap and boundaries are soft; some customers sit between two segments
K chosen by stability and usefulness, not by a single metric	Silhouette is highest at K = 2 and the elbow suggests K = 5, so K = 4 is a judgement supported by reproducibility
Experiments on samples of 6,000 to 100,000 customers	Model-selection metrics are estimates and vary with the sample; only the final fit uses all customers
No ground-truth labels	Quality can be assessed only by internal metrics, stability and business sense, not by accuracy
Segment names are business interpretations	K-Means does not learn names; they depend on the analyst's reading of the medians
Clusters 1 and 3 share the same median recency and frequency	They are distinguished mainly by spending level and unit price
Returns, lifetime and spending trend not used in the final model	Return behaviour appears only in profiling and does not shape the segments
Single snapshot with a fixed analysis date	Recency changes over time, so segments will drift and the model needs periodic refitting
Repeat-customer trend model was unstable	Segmentation of repeat customers by spending trend remains unresolved
No demographic, channel or product-category data	Segments describe purchase behaviour only, not who the customers are

10. Conclusion

This project built customer-level behavioural features from sales and return transactions and used them to segment 1,283,346 customers. After validating the features, removing redundancy and correcting skewness, several feature sets and cluster counts were compared. The highest silhouette scores favoured two clusters, which only separate active from inactive customers, so the

final choice was based on reproducibility: a lean set of five features (recency, sale frequency, monthly spending, discount rate and average unit price) with K = 4 gave a mean ARI of 0.904 across independent training samples and an ARI of 0.994 against a full-data run.

The four segments, Loyal High-Value (35.6%), Inactive Moderate-Spending (28.0%), Discount-Driven Regulars (16.1%) and Lapsed Low-Spending (20.3%), differ clearly in recency, frequency, spending, discount use and unit price, and they can support loyalty offers, win-back campaigns and targeted discounting. The main caveats are the moderate separation between segments (silhouette ≈ 0.24) and the fact that the names are interpretations. The main lesson is methodological: when the silhouette score does not discriminate between options, testing how reproducible a clustering is gives a more reliable basis for choosing the number of clusters.

11. References

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